LeasePulse Taipei

A Data Monetization System for Greater Taipei Rental Pricing Intelligence

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Big Data Systems — Final Project — Spring 2026

GitHub Repository: https://github.com/FettHsiao/bdfinal

Clone URL: https://github.com/FettHsiao/bdfinal.git

Live Demo: Local product demo: run `make api` (http://localhost:8000/) and `make dashboard` (http://localhost:8501/) — see repository README

Report generated on 2026-06-14 UTC

1. Executive Summary

LeasePulse Taipei is a B2B data product that transforms Taipei City open-data rental transactions into actionable pricing bands for a narrowly defined customer segment: independent landlords managing 2–15 units in Greater Taipei. The system downloads the official weekly real-price CSV (570 rental rows across 12 districts in our latest pull), computes district-level rent statistics in batch, and delivers recommendations through a Streamlit dashboard and FastAPI. Revenue comes from tiered subscriptions (NT$499/month Pro, NT$999/month API) justified by time saved on manual comparable-rent research and reduced vacancy risk.

2. Target Customer (Required Component 1)

Primary segment: Independent landlords in Taipei City and New Taipei City who self-manage 2–15 residential units and price vacancies without a dedicated analyst team.

Persona — Mr. Lin, 42:

- Owns 6 units across 大安區 and 板橋; works full-time in finance.
- Current workflow: browse 591 listings, ask agent friends, maintain an Excel sheet.
- Job-to-be-done: set a defensible listing rent within 48 hours of a tenant move-out.
- Why LeasePulse wins: district-level bands from Taipei open-data transactions, enriched quote breakdown (P25/median/P75), and exportable justification for tenants or co-owners.

Secondary segment: Micro property-management firms (5–20 employees) serving dozens of units; they would use the API tier to embed quotes in internal tools.

Attribute	Detail
Customer type	B2B — small business / prosumer landlord
Industry	Residential property management
Company size	1–15 units (primary); 50–200 units under management (secondary)
Status quo	Manual listing search, agent PDFs, spreadsheets
Wedge	Self-serve pricing confidence without enterprise contracts

3. Evidence of Demand (Required Component 2)

We validated demand using reproducible public evidence only — no private interviews. All collector logic lives under the `data/` package; `make run` executes every crawler, builds the public evidence report, ingests Taipei open data, and runs batch processing.

3.0 Data sources and reliability

Data source	Reliability	Limitations
Taipei City weekly real-price CSV (official open data)	High — government dataset with documented schema and weekly refresh	Delayed vs. listing platforms; addresses not stored in product DB
PTT rental-board crawl (last 2 years)	Medium — live public posts with source URLs and PII redaction	Forum sampling bias; not a formal market survey
Google Trends (24 months)	Medium — relative search-interest index from Google	0–100 index only; not absolute search volume
App Store RSS reviews (last 2 years)	Medium — public user reviews from Apple RSS feed	App-specific pain points; not landlord interviews
Competitor pricing pages (591 listing fees + analog benchmark)	Medium — public pages with manually verified prices and source URLs	Listing fees vary by plan; verify against cited pages

3.1 Google Trends search-interest signals (24 months)

We collected Google Trends interest scores for rent-pricing keywords over `2024-06-14 2026-06-14` in `TW`. The strongest keyword by mean interest was 租金行情. Trends values are relative scores (0–100), not absolute search volume, but they show sustained attention to rent-pricing topics.

3.2 App Store review signals (last 2 years)

We analyzed 150 public App Store reviews from rental/housing apps since `2024-06-14`. Average rating: 1.69. Top pain keywords: data_quality, trust, contact, ads, search_quality. These reviews provide qualitative demand and UX pain-point evidence without private interviews.

3.3 Public forum signal analysis (PTT crawl)

We crawled 118 public PTT rental threads using `data/collect_ptt_forum_signals.py`. Posts were limited to the last two years, filtered to Greater Taipei relevance, and redacted for common PII before analysis. 12% matched a pricing-question heuristic. Boards covered include Rent_apart, rent_tao, Rent_ya, home-sale. Recurring keywords include pricing, studio, comparable, elevator, landlord. The median rent figure mentioned in forum posts was approximately NT$7,817/month. Most discussed districts: 中山, 大同, 信義, 永和, 中和. PTT posts are treated as qualitative demand signals, not formal market statistics.

3.4 Competitor and analog pricing (live crawl)

We checked 4 public competitor/analog pricing pages via `data/collect_competitor_pricing.py` using sources listed in `data/sources/competitor_pricing_sources.json`. Products/pages: 591 出租刊登（普通方案）, 591 加值服務（定時更新）, 591 加值服務（精選推薦）, LeasePulse Taipei Pro（本專案定價）. Median monthly-like benchmark: NT$449. Manually verified benchmarks: 591 出租刊登（普通方案） (出租普通刊登) NT$400; 591 加值服務（定時更新） (定時更新) NT$150; 591 加值服務（精選推薦） (精選推薦區（出租）) NT$1,500; LeasePulse Taipei Pro（本專案定價） (Pro subscription) NT$499. Each record stores source URL, checked time, manual verified price when available, and extracted pricing snippets.

3.5 Live open-data acquisition process

Beyond qualitative validation, the product pipeline ingests real rental transactions from the Taipei City Data Platform weekly real-price report (`data.taipei`). The acquisition script `data/taipei_open_data.py` performs: (1) HTTPS download with curl/SSL fallback; (2) rental-row filtering on CASE_T/CASE_F; (3) unit normalization (TPRICE in 萬元, UPRICE in NTD/ping, SDATE ROC-date parsing); (4) export to `data/processed/transactions_ingest.csv`. Our latest run produced 570 cleaned rental records across 12 Taipei districts. Highest median-rent districts: 大安區 (NT$32,500/mo), 南港區 (NT$30,500/mo), 中正區 (NT$29,000/mo).

3.6 Willingness to invest (time, effort, money)

Investment type	Observed value	LeasePulse value proposition
Money	Listing / tool fees on incumbent platforms	Pro tier at NT$499; API at NT$999
Time	Manual listing search + spreadsheet work	Quote in < 30 seconds
Effort	Multi-tab listing search + Excel	Single dashboard + export
Risk cost	3+ weeks vacancy if overpriced	P25–P75 band reduces tail risk

4. Technical System Design

4.1 Data sources and ingestion

The system downloads the official Taipei City weekly real-price CSV through `data/taipei_open_data.py` and loads it via `data/ingest.py --fetch`. The open-data endpoint is hosted on the Taipei Data Platform. Each record includes district (行政區), building type (建物型態), area in ping, monthly rent (NTD), and transaction date parsed from ROC-format SDATE. Raw files are archived under `data/raw/`; cleaned ingest files under `data/processed/`. `make run` from the repository root runs public demand evidence (PTT, Google Trends, App Store reviews, competitor pages), open-data ingest, and batch processing.

4.2 Storage and processing

Structured facts land in PostgreSQL (SQLite locally) via SQLAlchemy models. A batch processor (`pipeline/processor.py`) computes per-district medians and 25th/75th percentile rent-per-ping bands using Pandas. We also reuse Homework 2's MapReduce K-Means implementation (`pipeline/mapreduce_kmeans.py`, adapted from `hw2/hw_r14921059/mapper.py` and `reducer.py`) to segment the market into budget/value/premium/luxury clusters on features (area_ping, rent_per_ping). At 100× scale, Pandas and K-Means map to Spark batch jobs over parquet files in object storage.

4.3 Delivery

Customers consume the product through two layers:

- Streamlit dashboard (`dashboard/app.py`): district rent charts, HW2 K-Means segments, and an interactive quote form. After submission, the quote view shows conservative/market/aggressive monthly rent, rent-per-ping, annual rent, a bar chart, sample-size/confidence metadata, and strategy notes.
- FastAPI REST layer (`api/main.py`): typed endpoints including `/health`, `/metrics/districts`, `/quote`, `/clusters`, and `/docs`. The root path `/` renders an HTML navigation page with clickable links to each endpoint so graders can explore the API without memorizing URLs.

• Enriched quote API: `/quote` returns P25/median/P75 monthly rent, rent-per-ping bands, annual rent estimates, pricing guidance labels, sample size, and last-updated timestamp — not just a single number.

4.4 Architecture diagram

Data Sources	→	Taipei open data	→	PostgreSQL	→	Pandas + HW2 K-Means
				↓		
		FastAPI	←	Aggregates	→	Streamlit Dashboard

Layer	Technology	Rationale
Ingestion	Python + requests/curl	Official Taipei CSV download
Storage	PostgreSQL / SQLite	Relational aggregates, indexed lookups
Processing	Pandas + HW2 MapReduce K-Means	District stats + market segments
API	FastAPI + Uvicorn	Typed REST delivery, easy deployment
UI	Streamlit + Plotly	Rapid SMB dashboard, demo-friendly
Ops	Docker Compose	Reproducible local and cloud deploy

4.5 Implementation snapshot

Artifact	Location	Purpose
Open-data fetch	data/taipei_open_data.py	Download & clean Taipei CSV
PTT forum crawl	data/collect_ptt_forum_signals.py	Public demand-evidence collection
Evidence report	data/collect_public_demand_evidence.py	Public demand evidence summary
Ingestion CLI	data/ingest.py --fetch	Load live data into DB
Batch pipeline	pipeline/processor.py	District metrics + K-Means
API home	GET /	Clickable endpoint navigation
Quote UI	dashboard/app.py	Expanded rent-band breakdown
Report generator	report/generate_report.py	Regenerate r14921059.pdf

4.6 Scalability and unit economics (optional)

At MVP (~570 live transactions, 50 paying users), estimated infra cost is under USD 25/month on a small VM plus managed Postgres. Gross margin at NT$499 × 50 ≈ NT$25k/month exceeds infra by an order of magnitude. At 100× data volume, batch processing shifts to Spark; API read load is served from precomputed aggregates, keeping query latency stable.

5. Go-to-Market Difficulties (Bonus Component 3)

• Trust and adoption: Landlords may trust agent relationships over an unknown SaaS dashboard. Mitigation: show transaction sample sizes, confidence scores, and side-by-side comparisons with their current manual method.

• Data acquisition cost: Government open data is free but delayed; listing platforms are fresher but license-bound. Over-reliance on scraped listings creates ToS and PDPA exposure.

- Legal / privacy: Even aggregate rental products must avoid exposing identifiable tenant or landlord records. Production requires PDPA review and clear retention policies.
- Cold start: Sparse districts yield wide bands and low confidence — early users in Neihu vs. Da'an may get uneven value. We surface confidence explicitly to set expectations.
- Competition: 591, HousePlus, or banks could bundle similar analytics. Our moat is narrow focus on micro-landlords, faster self-serve UX, and pricing aligned to their willingness to pay — not enterprise sales cycles.
- Unit economics: CAC via Facebook landlord groups is low, but churn rises if vacancies are rare (users pause subscriptions). Annual plans and vacancy-triggered alerts improve retention.

6. Business Model Summary

Free tier: read-only district snapshot. Pro (NT$499/mo): custom quotes, exports, email support. API (NT$999/mo): programmatic access for micro property managers. Optional one-off vacancy report (NT$299) acts as conversion hook. Primary acquisition: PTT/LINE landlord communities, NTU alumni networks, and agent referral rebates.

7. Repository and Reproducibility

Source code, configuration, and documentation are published at:

`https://github.com/FettHsiao/bdfinal`

Clone with:

`https://github.com/FettHsiao/bdfinal.git`

The repository includes ingestion scripts, batch pipeline, API, dashboard, Docker configuration, HW2 reference code, and demand-evidence reproduction steps in README.md.

Quick reproduction steps:

- `git clone https://github.com/FettHsiao/bdfinal.git && cd bdfinal`
- `python3 -m venv .venv && source .venv/bin/activate && pip install -e .`
- `make run # ptt + trends + reviews + competitors + evidence + ingest + process`
- `make api # terminal 1 → http://localhost:8000/`
- `make dashboard # terminal 2 → http://localhost:8501/`
- `make report # regenerate r14921059.pdf`

8. Conclusion

LeasePulse Taipei demonstrates that course-scale big-data tooling — live open-data ingestion, SQL storage, batch analytics, MapReduce-style segmentation, and API/dashboard delivery — can support a credible data monetization story when paired with rigorous customer definition and demand evidence. The defensible claim is not 'big data magic,' but a focused product grounded in verifiable Taipei City transactions that saves measurable landlord research time and reduces costly vacancy mistakes.